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Note: Live quiz MCQs also repeated in Exam (6, 7)



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# PAID VU LMS HANDLING by Mam Mehwish 03184148783

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## MTH603 (Mega File)

## Mid Term (Live Quiz)

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**1. For the power series, define R in the extended real numbers by 1/R = lim sup n→∞ $|a_n|^{1}/n$. In particular, R = ∞ if...**

- **lim sup n→∞ $|a_n|^{1}/n = 0$.**
- **lim sup n→∞ $|a_n|^{1}/n = \infty$.** ✅
- lim sup n→∞ $|a_n|^{1}/n = 1$.
- lim sup n→∞ $|a_n|^{1}/n = -\infty$.

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**2. For the power series; f(z) = ∑ (−1)ⁿ z^(2n+1) / (2n+1)!, then f'(z) =...**

- **∑ $(-1)^n z^{2n} / (2n)!$**
- **sin x = ∑ $(-1)^n x^{2n} / (2n)!$**
- ∑ $(-1)^n x^{2n+1} / (2n+1)!$ ✅
- ∑ $(-1)^n x^{2n+2} / (2n+1)!$

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**3. tan x is an odd function, its derivatives of even order are equal to...**

- **1 at $x_o = 0$.**
- π at $x_o = 0$.
- **0 at $x_o = 0$.** ✅
- tan x sec x at $x_o = 0$.

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**4. The power series expansion of f(x) = sin x about $x_o = 0$ is...**

- $\sum^{\infty} n=0\ x^{2n} / (2n)!$
- **$\sin x = \sum^{\infty} n=0\ (-1)^n\ x^{2n+1} / (2n+1)!$** ✅
- $\sum^{\infty} n=0\ (-1)^n\ x^{2n} / (2n)!$
- $\sum^{\infty} n=0\ x^{2n+1} / (2n+1)!$

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**5. If f(x) = sin x, then...**

- **f''(x) = sin x** ✅
- $f^{2m+1}(0) = 1,\ m \geq 0.$
- $f^{2m}(0) = 1.$
- $||f^{(k)}||_{(-\infty,\infty)} = 1,\ k \geq 0.$

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**6. If a function f(x), having the derivatives of all orders on some neighborhood N(0) of 0, be chosen first and the Taylor series $\sum^{\infty} n=0\ f^{(n)}(0) / n!\ x^n$ be constructed, then this power series will have f(x) as its sum on N(0).**

- **True** ✅
- False

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**7. Let f be defined by a power series with finite radius of convergence R. Then $\lim x \to (x_o+R)^-$ f(x) = $\sum^{\infty} n=0\ a_n\ R^n$ only if...**

- None of these.
- $\sum^{\infty} n=0\ (-1)^n\ a_n\ R^n$ converges.
- **$\sum^{\infty} n=0\ a_n\ R^n$ converges.** ✅
- $\sum^{\infty} n=0\ b_n\ R^n$ is uniformly convergent.

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**8. The infinite series expansion of cos x for $-\infty < x < \infty$ is given by...**

- $\sum_{n=0}^{\infty} (-1)^n x^{2n} / (n+1)!$
- $\sum_{n=0}^{\infty} (-1)^n x^{2n} / (2n)!$
- **$\sum_{n=0}^{\infty} (-1)^n x^{2n} / 2n!$** ✅
- None of these.

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